INSTRUMENT CLUSTER (Continued)

driver/passenger door ajar warning, the head/park lights-on reminder, and the key-in ignition reminder. For the remaining chime warning functions, the EMIC uses a combination of hard wired inputs, electronic message inputs received from other modules over the PCI data bus, and internal programming. (Refer to 8 - ELECTRICAL/CHIME/BUZZER - OPERATION).

The hard wired chime inputs to the EMIC may be diagnosed using conventional diagnostic methods. However, proper testing of the EMIC programming and the electronic chime request messages received by the EMIC over the PCI data bus requires the use of a DRBIII® scan tool. Refer to the appropriate diagnostic information.

ELECTRONIC PINION FACTOR

Each time the EMIC receives an electronic pinion factor request message from the Powertrain Control Module (PCM) over the PCI data bus, it provides a response to the PCM containing the appropriate electronic pinion factor information. Because there are numerous optional combinations of tire size, axle ratio, and transfer case type which are available for this vehicle, the electronic pinion factor needs to be considered for accurate processing of the vehicle speed and distance information. The instrument cluster stores the electronic pinion factor information and broadcasts it over the PCI data bus upon request by the PCM. Using this factor and the Vehicle Speed Sensor (VSS) input, the PCM then calculates the proper vehicle speed and distance information, and transmits the appropriate electronic vehicle speed messages over the PCI data bus for use by other electronic modules in the vehicle.

Proper testing of the PCI data bus and the electronic data bus message inputs to and outputs from the EMIC that control the electronic pinion factor function requires the use of a DRBIII® scan tool. Refer to the appropriate diagnostic information.

INTERIOR LIGHTING CONTROL

The EMIC contains an integral timer and logic circuit to perform both timer and control functions for the interior courtesy lamps. The EMIC uses hard wired inputs from the ignition switch, both door ajar switches on separate driver and passenger door ajar switch sense circuits, from the resistor multiplexed panel lamps dimmer circuitry of the left multi-function switch on the panel lamps dimmer signal circuit and its control logic to provide a battery current output to the courtesy lamps on a courtesy lamp feed circuit. The EMIC control provides a theater-type fade-to-off feature that will slowly dim the courtesy lamps about five seconds after both doors are closed.

The EMIC interior lighting control programming provides an illuminated entry/exit feature by monitoring the door ajar and ignition switch inputs. When a door is opened with the ignition switch in the Off position, the EMIC turns on the courtesy lamps. When the ignition switch is turned to the On or Start positions, the EMIC turns the courtesy lamps off immediately with no theater dimming. When the ignition switch is turned from the On position to the Off position, the EMIC turns on the interior lights for about ten seconds or until the ignition switch is again turned to the On or Start positions, whichever occurs first.

The EMIC interior lighting control programming also provides a battery saver feature (load shedding) for the interior lighting. Unless the engine is running, the EMIC will automatically turn off the interior lights if they are left on for more than about twenty minutes, regardless of the status of the ignition switch, door ajar switch, or left multi-function switch inputs to the cluster.

The hard wired inputs and output of the EMIC interior lighting control may be diagnosed using conventional diagnostic methods; however, there are no other diagnostic tools available for the EMIC timer and logic circuitry. If the input and output components and circuits of the interior lighting system test OK, but the system fails to operate as designed, the EMIC must be replaced.

REAR WINDOW DEFOGGER CONTROL

The EMIC contains an integral timer and logic circuit to perform the rear window defogger timer and control functions for the optional rear window defogger system. The EMIC uses a hard wired input from the rear window defogger switch on the rear window defogger switch sense circuit and its control logic to determine the correct output to the rear window defogger relay. The EMIC controls the ground path of the rear window defogger relay control coil through an output on the rear window defogger relay control circuit.

The EMIC is programmed to interpret each momentary ground signal it receives on the rear window defogger switch sense circuit as a request to change the current state of the output on the rear window defogger relay control circuit. Therefore, with the ignition switch in the On position, the first ground input on the rear window defogger switch sense circuit turns the system On, the second ground input turns the system Off, and so forth. Once the rear window defogger system has been turned On, it can be turned off manually by depressing the rear window defogger switch a second time or by turning the ignition switch to the Off position.